

Osaid Khan

+1 (945) 304-5781 | khanosaid726@gmail.com | www.linkedin.com/in/-osaid-khan/ | github.com/ozzyozbourne

EDUCATION

Pace University

Master of Science (MS) in Computer Science (GPA - 4.0/4.0)

New York City, NY

Jan 2024 - Dec 2025

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology (B.Tech) in Computer Science

Noida, India

Jul 2016 - Sep 2020

TECHNICAL SKILLS

AI & Machine Learning: LLM Applications, RAG (Retrieval-Augmented Generation), LangChain, LangGraph, LangSmith, VectorDB, PyTorch, Gemini, GPT, CUDA, vLLM

Backend & Languages: Node.js (Express), JavaScript/TypeScript, Java, Python, Rust, Spring Boot, FastAPI, RabbitMQ, Kafka

Cloud & DevOps: AWS (EC2, S3, Lambda, Step Functions, CloudWatch), Docker, Kubernetes, CI/CD, Google Cloud, Redis, PostgreSQL, MongoDB

Others: Security by Design, SDLC Security, Git, nx, vite, rush

PROFESSIONAL EXPERIENCE

Software Engineer

Sperse

Phoenix, AZ

Feb 2026 - present

- Architected a **concurrent multi-agent** orchestration system using **LangChain** and **LangGraph**, enabling scalable tool execution and reliable coordination across complex, **enterprise-grade workflows**
- Developed and delivered an MCP AI interaction layer that translates natural-language intents into **safe, validated tool calls** across **1000+ platform endpoints**, ensuring **Security by Design** for automated actions
- Optimized prompt routing and **LLM agent** concurrency to maximize **KV-cache hit rate**, reducing end-to-end latency by **60%** and lowering API costs by **50%** while maintaining high-quality outputs
- Built **generative code solutions** (API-aware templates and SDK patterns) to accelerate developer integration with **Node.js** platform **APIs** and improve the correctness of implementation cycles
- Established a robust **observability layer** using **LangSmith** for **tracing, structured logs**, and **evaluation hooks** to monitor **LLM agent** behavior, debug failures, and track performance metrics over time

AI Researcher

Pace Artificial Intelligence Lab

New York City, NY

Sep 2025 - Dec 2025

- Conducted research on **StyleGAN** and **Stable Diffusion** using **PyTorch** and **CUDA** on HPC clusters, managing the full **ML lifecycle** from model training and fine-tuning with **LORA** to efficient model serving
- Utilized **vLLM** for high-performance model inference with **Hugging Face** deployment, ensuring secure and scalable delivery of **LLM applications**
- Led technical workshops on GPU programming and building **AI Agents** with **LangChain** and **LangGraph** for **+50** students, promoting engineering best practices and **mentorship** within the lab

Software Engineer

Qualitest

Noida, India

Feb 2021 - Dec 2023

- Built an internal support **Ambient Agent** using **LangChain** and **RAG** with **PostgreSQL** vector store, enabling natural language queries over documentation and reducing ticket resolution time by **20%**
- Designed and developed a cloud-agnostic Storage Service using **Node.js** and **Python** that handles file operations across **S3, Google Cloud Storage**, and **MinIO** via secure, expiring signed URLs
- Developed complex backend workflows using **AWS Step Functions** and **Kafka** to handle long-running data import jobs and prevent **Lambda** timeouts, ensuring high **availability** and **scalability**
- Contributed to an Identity & Access Management platform using **React**, **Spring Boot**, and **AWS**, supporting thousands of users across 30+ applications with a focus on **security tooling** in the **SDLC**

PROJECTS

Dark Factory - Orchestrator for AI coding CLIs | *Crossterm, Ratatui, Y-crdt*

github.com/ozzyozbourne/dark-factory

- Engineered an **event-driven** orchestrator for **agentic loops** using **Y-crdt** to eliminate polling overhead and enable real-time, distributed session broadcasting

- Implemented secure CLI IO ownership via **PTY**, validating and logging every command before execution by the bash shell to ensure **safe tool calls**